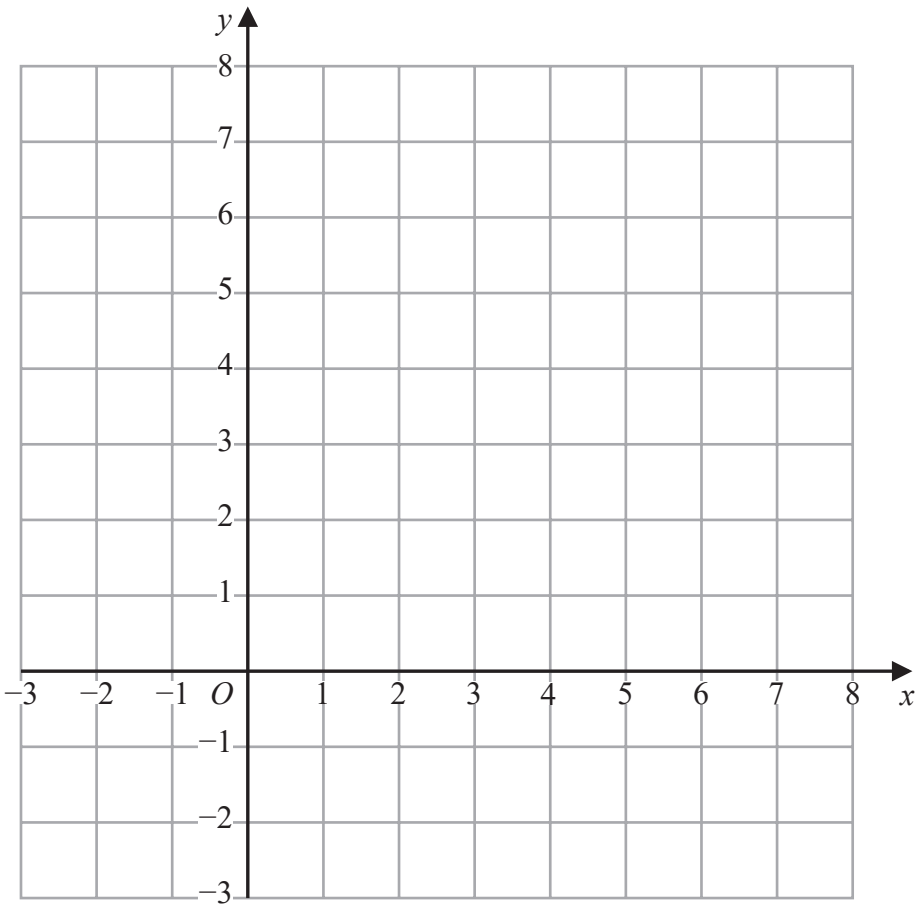


13 (a) On the grid show, by shading, the region that satisfies all these inequalities.

$x \geq 0$ $x \leq 2$ $y \leq x + 3$ $2x + 3y \geq 6$

Label the region **R**.

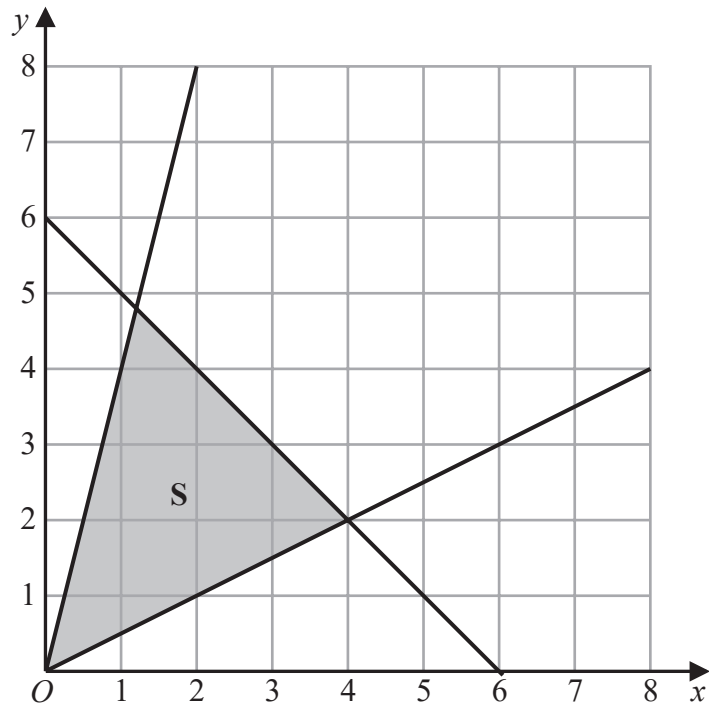


(b) The diagram below shows the region **S** that satisfies the inequalities

$y \leq 4x$

$y \geq \frac{1}{2}x$

$x + y \leq 6$



Geoffrey says that the point with coordinates (2, 4) does not satisfy all the inequalities because it does not lie in the shaded region.

Is Geoffrey correct?
You must give a reason for your answer.

(1)